

Machine Learning for Site of Origin Classification in Idiopathic Ventricular Arrhythmias

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I. INTRODUCTION

Idiopathic Ventricular Arrhythmias (VAs) are characterized by abnormal electrical pulses in patients presenting no structural heart disease. They manifest as premature ventricular contractions (PVCs) or ventricular tachycardias (VTs). These arrhythmias predominantly originate within the right ventricular outflow tract (RVOT) and left ventricular outflow tract (LVOT). More specifically within these tracts, the exact Site of Origin (SOO) spans distinct anatomical micro-structures, including the ventricular septum, the free wall, the left and right coronary cusps and the LVOT summit area [1].

Radiofrequency Catheter Ablation (RFA) represents the definitive therapeutic approach to eliminate these arrhythmogenic SOO via localized thermal endocardial scarring. The clinical success of RFA is dependent on precise pre-procedural mapping. Clinicians traditionally rely on the non-invasive analysis of the standard 12-lead surface electrocardiogram (ECG), where the morphological configuration of the QRS complex mirrors the initial spatial pathway of ventricular depolarization [2].

Current clinical practice of localizing the SOO via ECG suffers from severe limitations. It relies heavily on visual analysis and manual scoring indexes such as the Weighted Hybrid Score. These methods besides slowing down clinical workflows during pre-procedural evaluation suffer from high inter-observer variability. Therefore distinguishing between closely intertwined, adjacent spatial micro-structures generates substantial inter-observer divergence. This impacts the success of treatments that are highly dependent on the correct localization of the SOO, such as RFA [3]. Furthermore, when pre-procedural localization is imprecise, medical teams are forced to rely on prolonged, invasive intra-cardiac mapping during the intervention. This negatively affects the patient as the operation times increase as well as the radiation exposure. Moreover, these types of invasive procedures have a lower first-pass ablation success rate.

Advanced computational frameworks offer a data-driven path to address these challenges. By extracting high-dimensional spatiotemporal features from raw or digital ECG waveforms, machine learning models can uncover subtle morphological

patterns. Furthermore, such frameworks could include clinical variables to further support the SOO classification.

This study presents an end-to-end, fully autonomous localization framework that handles data processing, feature curation and algorithmic modeling across three classification complexities: binary classification, intermediate multi-class classification and fine-grained SOO classification. The different models for classification were tested against a completely isolated external validation cohort (from the Teknon dataset), to evaluate the readiness of deep learning systems for real world clinical applications. To ensure reproducibility and transparency, the complete implementation, preprocessing pipelines, and trained models are publicly available in a GitHub repository (Appendix A).

Results: Traditional architectures overfitted heavily on raw high-dimensional representations. However, transitioning to an engineered morphological feature selection pipeline stabilized classical models yielding a top intermediate multi-class balanced accuracy of 59.79% via Gradient Boosting. On the other hand, an end to end 1D Convolutional Neural Network (CNN) architecture with Data Augmentation demonstrated optimal binary performance archiving 76.67% balance accuracy. Severe class scarcity and minority-class oversampling constraints remain the primary bottlenecks for fine-grained localization.

II. METHODOLOGY

A. Datasets

Three datasets were used for model development. DS-43 contains 31 12-lead ECG recordings from Hospital Clínic (Barcelona), and DS-334 is an open-source database of 334 patients published by Zheng et al. [4]. Both are provided as pre-segmented QRS complexes with binary RVOT/LVOT labels. The primary dataset is DS-181, collected at Hospital Teknon (Barcelona), which is the only one providing full 12-lead ECG recordings alongside patient-level clinical data for 181 patients. DS-181 is used for all evaluations, with the final 20% reserved as a held-out test set that is never accessed during development of the models.

A fourth dataset of 2,496 simulated ECGs (DS-2496) was available but ultimately excluded based on empirical evaluation on the development holdout, where including simulated data consistently degraded performance on real Teknon recordings. While simulation-based approaches have shown promise in prior work [5], in our setting the gap between physics-based synthetic signals and real clinical acquisitions appeared to hurt rather than help generalization.

The class distribution is clearly imbalanced across all tasks. In the binary setting, RVOT cases represent approximately two-thirds of the Teknon cohort (92 vs 48 patients). The imbalance becomes more severe in Tasks 2 and 3: within the LVOT group, aortic cusp origins (RCC, LCC, Commissure) have fewer than 15 patients each, while RVOT Septum alone accounts for nearly half of all cases. This directly motivates the class balancing strategies described in Section III.

B. SOO label standardization

The Teknon SOO labels were stored as free-text clinical annotations, with approximately 90 unique values. These were standardized into 7 anatomical classes: RVOT Septum, RVOT Free Wall (RFW), RCC, LCC, Commissure, LVOT Summit, and LVOT Subvalvular. A rule-based mapper searched for anatomical keywords in each label and assigned the first matching class. Cases outside the outflow tract, such as fascicular or lateral annulus origins, were excluded, leaving 140 patients for model development.

C. Signal representation and ECG preprocessing

Rather than using the full 4-beat ECG recording, each patient was represented through the QRS complex of the PVC beat. This choice is motivated by three complementary reasons: first, the QRS reflects ventricular depolarization and is the primary signal used in the SOO literature [6,7]; second, it makes Teknon compatible with DS-43 and DS-334, which are already provided as QRS complexes; third, it yields a compact input that reduces overfitting risk given the small cohort size.

The preprocessing pipeline starts by resampling each Teknon ECG from 1000 Hz to 250 Hz, followed by a bandpass filter (0.5–100 Hz) to remove baseline wander and high-frequency noise. A 5-model U-Net ensemble [8] then performs P-QRS-T segmentation, from which the last-beat QRS (corresponding to the PVC) is extracted. Each QRS is resampled to 120 samples per lead using FFT-based resampling and amplitude-normalized by dividing by the maximum absolute value, making recordings comparable across patients regardless of electrode placement or gain differences.

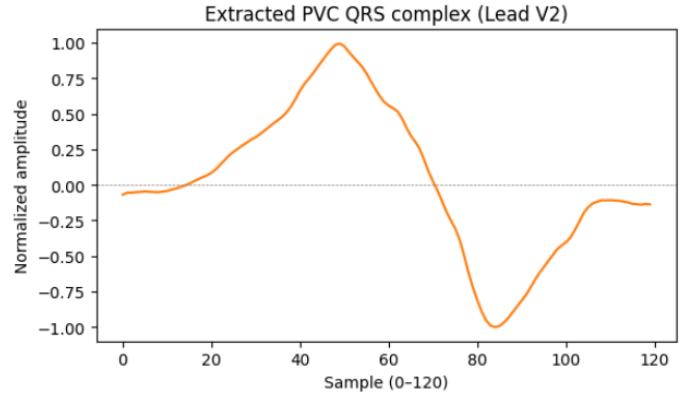


Fig. 2. Extracted and amplitude-normalized PVC QRS complex for lead V2 (120 samples), corresponding to the last beat of the ECG recording.

D. QRS Feature Extraction

For classical models, the raw QRS is not used directly. Instead, a set of morphological descriptors is computed per lead: amplitude statistics (mean, std, max, min, RMS, peak-to-peak), absolute and signed area, R/S ratio, R-S amplitude difference, R and S peak positions, maximum upward and downward slopes, zero-crossing rate, early absolute area, and 24 temporal bins capturing the shape of the signal over time. This produces a large initial candidate set. A two-stage feature selection then reduces it: first, feature relevance is assessed using both ExtraTrees importance scores and mutual information with the target, retaining features that rank highly in either criterion; second, highly correlated features (Pearson $r > 0.96$) are removed to avoid redundancy. The final feature set contains between 18 and 48 features depending on the task and dataset.

E. Clinical Feature Selection and Preprocessing

Following the clinical baseline in Bocanegra-Pérez et al. [6], we considered Age, Sex, HTA, and PVC transition as candidate features. Age and PVC transition showed significant associations ($p = 0.025$ and $p = 0.005$ respectively); Sex and HTA did not reach significance individually ($p > 0.3$), but were retained given their presence in literature such as the Penela et al. [3] weighted hybrid score. Beyond that, univariate statistical tests only capture marginal associations and may miss contributions that emerge in combination with other features, which the models used here are able to exploit during training [9].

The reference paper additionally uses V3 lead amplitude as a clinical feature [6]. In this pipeline, this is not included as an explicit clinical variable because it is already captured

implicitly by the QRS morphological descriptors (specifically the per-lead amplitude statistics and temporal bins computed for V3) in a richer form than a single scalar value.

Age was imputed with the median and standardized. PVC transition was ordinally encoded preserving the clinical V1→V6 ordering and imputed with the most frequent value. Sex and HTA were one-hot encoded with the first category dropped.

F. Cross-validation

To evaluate the different models in a robust manner, a 5-fold stratified cross-validation scheme was employed. The dataset was partitioned into five folds using stratified sampling, ensuring that the class proportions present in the full dataset were preserved within each fold. This was a critical consideration given the inherent class imbalance in the SOO categories.

For each of the five iterations, one fold was held out as the validation set while the remaining four folds constituted the training set. To mitigate the limited size of the data and improve model generalization, a data augmentation procedure was applied where augmented samples were generated through the addition of Gaussian noise ($\sigma = 0.008$) and random amplitude scaling (range $\pm 10\%$). Critically, augmentation was performed after the train-validation split at each fold, preventing data leakage that could arise from validating on synthetic variants of training samples.

A fresh instance of each model architecture was initialized for each fold (with fold-specific random seeds for reproducibility), and trained using the Adam optimizer ($\text{lr}=1\text{e-}3$, $\text{weight_decay}=1\text{e-}4$) with a step-decay learning rate schedule (halving every 20 epochs).

For each fold, performance was quantified using standard classification accuracy and balanced accuracy (the latter providing a more informative metric under class imbalance as it averages per-class recall).

G. CNN training setup:

As a novel contribution, a 1D convolutional neural network was trained directly on the raw QRS signal, using each of the 12 leads as an independent input channel. The architecture consists of two convolutional blocks, global average pooling, and a fully connected classifier with Dropout ($p=0.5$). Inverse-frequency class weighting was applied to the cross-entropy loss function to compensate for class imbalance. Training used the Adam optimizer ($\text{lr}=1\text{e-}3$,

$\text{weight_decay}=1\text{e-}4$) with StepLR scheduling (step=20, $\gamma=0.5$). To control overfitting, early stopping with patience of 15 epochs was used alongside the dropout layer described above.

H. Success metrics

The evaluation of the model was done with two main metrics: *Accuracy* and *Balanced Accuracy*.

- **Accuracy (Eq. 1):** this metric measures the proportion of all classifications that were correct, whether positive or negative [10]. It is defined as:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (\text{Eq. 1})$$

Where TP is the number of True Positives, TN is the number of True Negatives, FP is the number of False Positives and FN is the number of False Negatives.

- **Balanced Accuracy (Eq. 2):** computes the average recall across all classes. This metric is particularly relevant in this study because the SOO classes are highly imbalanced [10].

$$\text{Balance Accuracy} = \frac{TPR+TNR}{2} \quad (\text{Eq. 2})$$

where TPR is the true positive rate and TNR is the true negative rate.

Confusion matrices were also examined to identify the most frequent misclassifications between anatomical groups.

III. EXPERIMENTS

A. Multi-Models classification

Five different models were evaluated: *Logistic Regression*, *Support Vector Machine*, *Random Forest*, *Random Forest Extra Trees* and *Gradient Boosting Trees*.

- **Logistic Regression:** It estimates the probability that a patient belongs to each class by applying a sigmoid function to a linear combination of the input features. L2 regularization is used to prevent the model from fitting to noise in the high-dimensional QRS feature space since it penalizes large coefficient values.
- **Support Vector Machine:** It calculates the hyperplane in the feature space that separates the classes with the largest possible margin. A linear kernel is used which means that the decision boundary is a straight hyperplane in the original feature space.
- **Random Forest:** It builds 100 decision trees and each one is trained in a random subset of the data and

a random subset of the features. The final prediction is made by majority voting across all trees. In this project two Random Forests were tested, a baseline model and one with extra trees.

- **Gradient Boosting Trees:** Similar to *Random Forest* it combines multiple decision trees but the construction of the trees is sequential with each new tree trained specifically to correct the errors made by the previous tree.

To ensure a robust evaluation, 20% of the Teknon dataset was strictly reserved as an external testing set. The DS-334 dataset, the Hospital Clínic dataset, and the remaining 80% of the Teknon dataset, formed a centralized training pool. From this pool, five different data combinations were generated to train the models. For each combination, an 80/20 split was applied: 80% was used for training, and 20% was held back for internal validation. Once trained, the models were evaluated against both their respective internal validation sets and the independent external test set. This methodology was applied to both the raw signals and the signals following feature selection. Ultimately, the model exhibiting the best performance was selected for each classification task.

B. CNN models

For Task 1, clinically valid data augmentation was applied: Gaussian noise ($\sigma=0.008$) to simulate electrode measurement variability, and amplitude scaling ($\pm 10\%$) to simulate inter-patient recording differences. Both perturbations were chosen to preserve QRS morphology, as shape-altering transforms such as temporal shifts or polarity flips would distort the diagnostic signal. Two training configurations were evaluated: full pool (488 samples) and Teknon only (112 samples), both with the same augmentation strategy, to assess the benefit of incorporating external datasets. For Task 2, where only Teknon is available and the class imbalance is more severe, selective oversampling was additionally used: class-specific augmentation factors (RVOT $\times 3$, LVOT $\times 6$, Cusps $\times 15$) combined with inverse-frequency class weighting.

IV. RESULTS

A. Classical Models with Raw QRS Representation

In the first stage, classical classifiers were trained directly on the raw QRS vectors, without morphology-based feature engineering. Each patient was represented by 1,440 ECG values, corresponding to 12 leads and 120 temporal samples per lead, together with the selected clinical variables. In this first stage the Random Forest with extra trees was the model selected for the three classification tasks.

Although the internal balanced tests produced high scores, performance decreased substantially when the models were evaluated on unseen Teknon patients. This can clearly be seen in the high accuracy scores obtained from the internal dataset were 92.5%, 94.44% and 93.86% for the binary task, the intermediate and the fine-grained respectively. These scores drop significantly when evaluating on the Teknon-holdout set, most notably the fine-grained score drops to a 10.71%. This reveals a marked generalization gap and motivates the subsequent feature-engineering pipeline.

B. Classical Models with Feature Selection

To reduce noise and improve generalization, a morphology-based feature-selection pipeline was implemented. The QRS format was homogenized across datasets and clinically meaningful descriptors were extracted. Feature selection was performed independently for each of the five balanced datasets. In practice, the selected feature sets ranged from 18 to 23 features for Task 1, and from 37 to 38 features for Tasks 2 and 3, reflecting the increased discriminative complexity of finer-grained classification. Patient-level separation was performed before oversampling to prevent data leakage. The best performing model for the binary task was the Random Forest, for the intermediate task was the Gradient Boosting and for the fine-grained classification was the Random Forest with extra trees.

TABLE I. Internal vs. External Validation Metrics for Outflow Tract Classification.

Task	Internal evaluation			Teknon-only holdout	
	CV	Acc.	Bal. Acc.	Acc.	Bal. Acc.
Binary	86.25%	85%	85%	53.57%	52.78%
Intermediate	94.43%	75%	75%	75%	59.79%
Fine-grained	85.1%	66.67%	66.67%	42.86%	25.42%

As shown in Table I, feature selection reduced the gap in scores between the internal test and the external one. For the binary task, the model achieved a best internal accuracy and balanced accuracy of 85%, which decreased to 53.57% accuracy and 52.78% balanced accuracy on the external Teknon-only holdout dataset. This smaller drop indicates a more stable training phase with less prominent internal overfitting.

In the intermediate multi-class task, the model recorded an internal accuracy and balanced accuracy of 75%; upon external evaluation, it yielded a raw accuracy of 75% alongside a noticeably lower balanced accuracy of 59.79%. This discrepancy indicates that while overall raw performance remains stable, the model struggles with class imbalance

resulting in significantly poorer performance on minority classes, which yields a lower balance accuracy.

Ultimately, the fine-grained task highlights the critical boundaries of the framework. The model's best internal metrics reached 66.67%, but plummeted to an external accuracy of 42.86% and a restricted balanced accuracy of 25.42% on the holdout cohort. This final decay underscores that the system remains profoundly limited by the severe data scarcity characteristic of the smallest anatomical sub-classes.

C. CNN Models for Binary Classification

A 1D-CNN was trained directly on the temporal QRS signal. Among the four evaluated configurations, the best binary result was achieved by the CNN trained with Data Augmentation, reaching 78.57% accuracy and 76.87% balanced accuracy. From Table II, the worst performing models are the ones trained with data exclusively from Teknon achieving an accuracy of 67.86% and 57.14% for the baseline model and the Data Augmentation respectively.

TABLE II. Performance evaluation on the hold-out Teknon test of 1D-CNN configurations for binary classification.

CNN configuration	Accuracy	Bal. Acc.
CNN baseline	75.00%	76.11%
CNN Teknon-only	67.86%	66.11%
CNN + augmentation	78.57%	76.67%
CNN + aug., Teknon-only	57.14%	53.33%

From the confusion matrix reported in Appendix B Fig B1.1, the selected model (CNN + augmentation) correctly classified 15 of 18 RVOT patients and 7 of 10 LVOT patients. This shows that in total only 6 patients were misdiagnosed. The corresponding recalls were 0.83 for RVOT and 0.70 for LVOT.

D. CNN Model for Intermediate Classification

For the intermediate task, selective oversampling was used to combat the scarcity of some of the classes. This model obtained an accuracy of 57.14% and a balanced accuracy of 44.71% on the Teknon hold-out test. Therefore, this CNN variant did not outperform the feature-selected Random Forest, reported in section B.

The confusion matrix for this CNN configuration is reported in Appendix B, Fig. B1.2. This matrix showcases that the highest performance is achieved for the RVOT class, whereas significant difficulties arise in the classification of the RCC/LCC/Commissure class. The latter represents the most underrepresented class within the dataset. When compared to the confusion matrix of the Random Forest model (Appendix B, Fig. B1.3), it can be observed that although both models

struggle with the same underrepresented class, the Random Forest approach demonstrates a superior classification accuracy specifically for the RVOT and LVOT cases.

E. Fine-Grained Classification

Fine-grained seven-class classification remained the most challenging task. Feature selection improved the Teknon-only balanced accuracy from 15.00% to 25.42%, but the result remained unstable.

Several anatomical classes contained very few independent Teknon patients. Oversampling increased the number of training samples but could not generate new clinical variability. Therefore, fine-grained predictions should be considered exploratory and require additional patient data before reliable validation.

The confusion matrix for the Random Forest classifier with Extra Trees, following feature selection, is reported in Appendix B, Fig. B1.4. While the RVOT and RFW classes exhibited the highest classification accuracy, performance became noticeably inconsistent for the underrepresented classes.

F. Final Model Selection

Model selection was based on Teknon-only holdout performance rather than internal balanced-test scores.

The selected binary model was the Teknon-only CNN with Data Augmentation. For the intermediate task, the best option was the feature-selected Gradient Boost. And finally, the fine-grained classification remains a proof of concept due to class scarcity.

G. SHAP-Based Interpretability Analysis

SHAP analysis was used to assess the contribution of each ECG lead to the CNN predictions. The corresponding lead-importance plots are provided in Appendix B.

For the binary RVOT/LVOT task, aVL showed the highest mean absolute SHAP value for both classes, followed by III and aVR. This suggests that limb-lead morphology played an important role in distinguishing right- and left-sided outflow tract origins (Appendix B, Fig B2.1-B2.2).

For the intermediate task, the most relevant leads varied across anatomical groups. V4 was the most influential lead for RVOT Septum/RFW, V1 for RCC/LCC/Commissure, and V6, followed by V5 and V4, for LVOT Summit/Subvalvular. These results indicate that precordial leads became more relevant when differentiating more specific anatomical regions (Appendix B, Fig B2.3-B2.4).

V. CONCLUSION & DISCUSSION

Automating the localization of the Site of Origin (SOO) in idiopathic ventricular arrhythmias via machine learning frameworks offers a promising avenue to eliminate inter-observer variability and optimize workflows. However transitioning these computational models from controlled internal validation environments to real-world clinical scenarios remains a challenge.

The primary finding of the investigation is the identification of a pronounced generalization gap that emerges as the specificity of the classification task increases. While models trained on raw representations (14440 unreduced features) yielded deceptively optimistic internal cross validation accuracy scores exceeding 92%, their performance failed when evaluated on the independent external validation cohort (20% of the Teknon dataset). This phenomenon shows that the models learned characteristics particular to the training datasets rather than capturing electrophysiological morphology during ventricular depolarization.

To mitigate this overfitting, the implementation of an end-to-end 1D Convolutional Neural Network architecture integrated with severe regularization techniques and Data Augmentation proved to be the most robust strategy for the binary task. This configuration achieved a stable balanced accuracy of 76.67% on the external holdout cohort, maintaining the best balance in accuracy between classes.

On the other hand, for the intermediate multi-class task and engineered morphology-driven feature selection pipeline combined with a Random Forest classifier successfully, obtained the best score for performance out of the other models scoring a 59.79% in the balance accuracy on completely unseen patient profiles.

When comparing the project's binary classification results with the ones reported from literature, the ones obtained are lower than those reported by Bocanegra-Pérez et al. [6], whose best supervised model achieved 89% accuracy and 86% macro-average sensitivity by combining ECG-derived information with clinical variables. The project developed omitted clinical variables as they worsened the prediction of the models as the other datasets used (Hospital Clinic and DS-334) did not include such variables. That is why a direct numerical comparison should be interpreted cautiously, since the validation strategy, feature representation and dataset composition differ between both studies.

Nevertheless, both approaches support the relevance of QRS morphology for RVOT/LVOT discrimination. While Bocanegra-Pérez et al. highlighted the contribution of

precordial leads V1–V4, particularly around the R/S transition and the initial QRS segment in V2, our SHAP analysis identified a stronger contribution from limb leads, especially aVL, III and aVR. These differences suggest that the most informative ECG patterns may depend on the modeling strategy and the characteristics of the evaluated cohort.

Despite these findings, the drop in performance from binary to fine-grained classification highlights the primary failure mode of this study: extreme data scarcity within specific micro-structures. As shown in the Teknon class distribution, certain classes such as commissure and LCC have too few patients to train a robust multi-class model. Models for the fine-grain classification obtained an external balance accuracy of 25.42%. This means that the system misclassifies rare anatomical variants of the majority classes and therefore introduces a diagnostic bias. Therefore, the model is unacceptable for real-world radiofrequency catheter ablation planning.

As mentioned previously, the main limitation of this project is the scarcity of the dataset as well as the extremely unbalanced dataset. For this reason, future work could focus on revising the standardization of the labels of the SOO to ensure it was done correctly. Moreover, the inclusion of other datasets that mimic really could be beneficial for the training of the models. As in this project we have seen that the inclusion of the simulated dataset worsen the performance of the models as it generates perfect signals that do not reflect reality.

Additionally, the oversampling introduced to mitigate class imbalance merely duplicated rare patient profiles. Consequently, these transformations failed to generate true, new clinical variability, limiting the model's ability to learn representative features. This meant that the model learned minor variations of identical waveforms that lead to a good performance in the internal validation but ultimately fail the external validation.

Future work for this project could include the expansion of the physiological window integrating the T-wave interval, to be able to capture a more complete electrophysiological profile. This would allow the model to view the ventricular repolarization dynamics, this would be beneficial as electrical recovery pathways vary significantly between the right and left ventricular outflow tracts. Therefore, this addition could help solve ambiguities and improve discrimination between adjacent SOO.

Finally, to address the severe class scarcity and anatomical imbalance that cause the multi-models to fail Prototypical Networks design for few-shot metric learning could be applied [11]. The network would be trained to project raw ECG signal

into a lower-dimensional embedding space where it computes an average patient prototyper for each anatomical microstructure. New, unseen patient waveforms will be classified based on their metric distance to these prototypes.

In conclusion, this study serves as a proof of concept for the automated site of origin localization in idiopathic ventricular arrhythmias. The project findings expose the difficulty of employing data-driven models within biological domains. The metrics achieved on the completely isolated external holdout cohort were 76.67% for the binary classification and 59.79% for the intermediate classification. The fine grained-model ultimately failed due to data scarcity.

While this project clearly maps the structural limits of traditional machine learning, expanding the physiological window to capture repolarization dynamics and transitioning toward few-shot metric frameworks represent the necessary path forward to safely navigate the rare anatomical variants of the human heart

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CNN variant for Task 2, although the feature-selected Random Forest achieved the best overall intermediate performance.

APPENDIX

A. GITHUB REPOSITORY

The following link contains the projects github repository:
<https://github.com/u199773/Ciencia-de-dades-i-models-computacionals-en-biomedicina-.git>

B. SUPPLEMENTARY FIGURES

This appendix contains the confusion matrices and SHAP lead-importance plots associated with the selected experiments.

B2. CONFUSION MATRICES

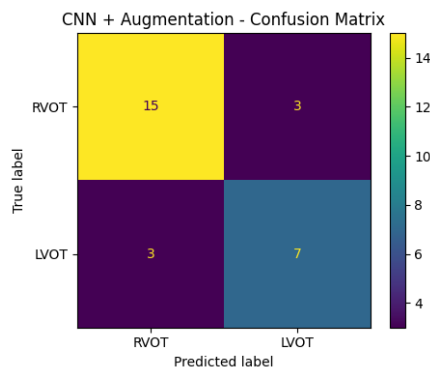


Fig. B2.1. Confusion matrix of the selected CNN with Data Augmentation for binary RVOT/LVOT classification on the Teknon hold-out test.

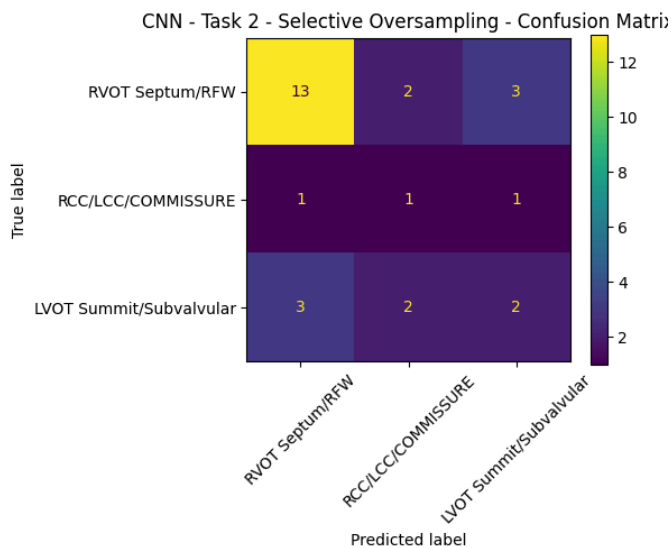


Fig. B1.2. Confusion matrix of the CNN with selective oversampling for intermediate three-class classification. This was the strongest

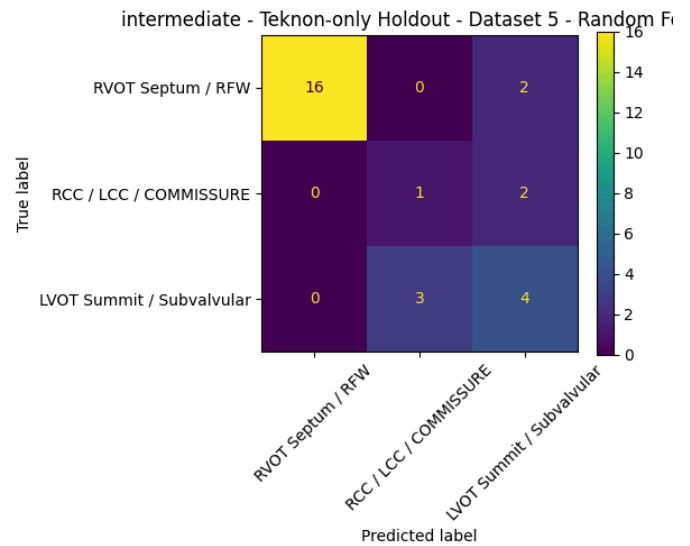


Fig. B1.3. Confusion matrix of the feature-selected Random Forest on the Teknon hold-out test for the intermediate classification task.

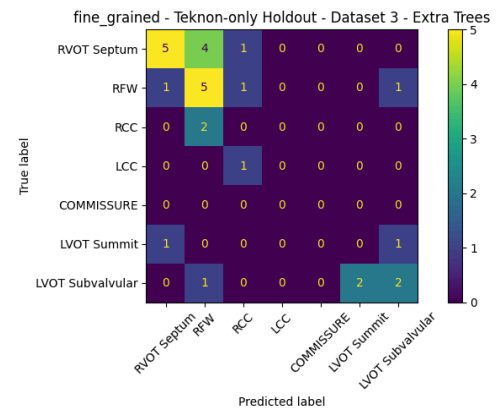


Fig. B1.4. Confusion matrix of the feature-selected Random Forest with extra trees on the Teknon hold-out test for the fine-grained task.

B. SHAP Lead-Importance Analysis

Mean absolute SHAP values represent the average contribution of each ECG lead to the model predictions.

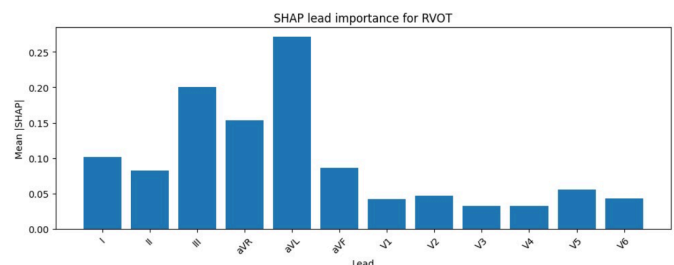


Fig. B2.1. SHAP lead importance for RVOT classification in the binary CNN model.

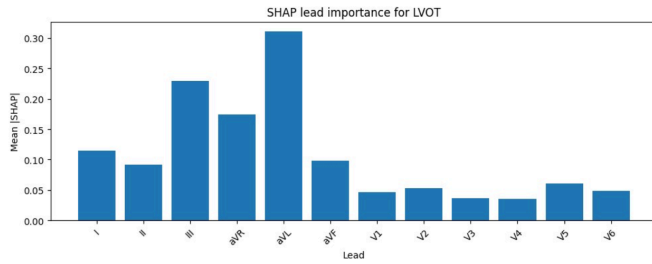


Fig. B2.2. SHAP lead importance for LVOT classification in the binary CNN model.

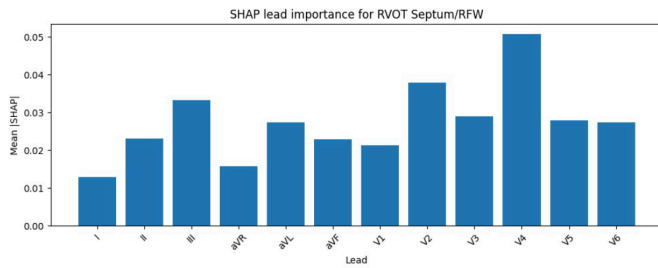


Fig. B2.3. SHAP lead importance for RVOT Septum/RFW classification in the intermediate CNN model.

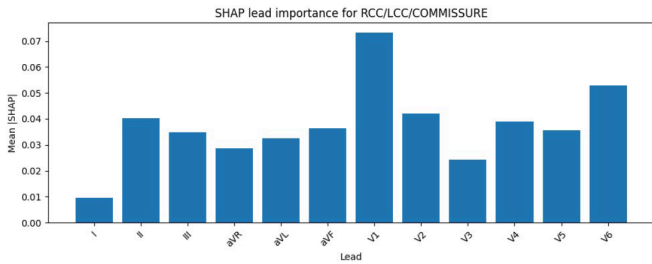


Fig. B2.4. SHAP lead importance for RCC/LCC/Commissure classification in the intermediate CNN model.

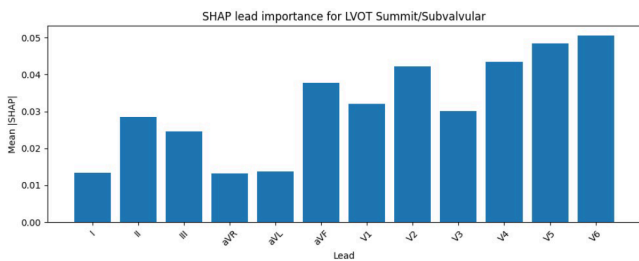


Fig. B2.5. SHAP lead importance for LVOT Summit/Subvalvular classification in the intermediate CNN model.